



Renewable Energy & Energy Efficiency

Introduction to Renewable Energy Resources — India Focus

INDIA'S CLEAN ENERGY JOURNEY

India's Energy Challenge

⚡ Surging Demand

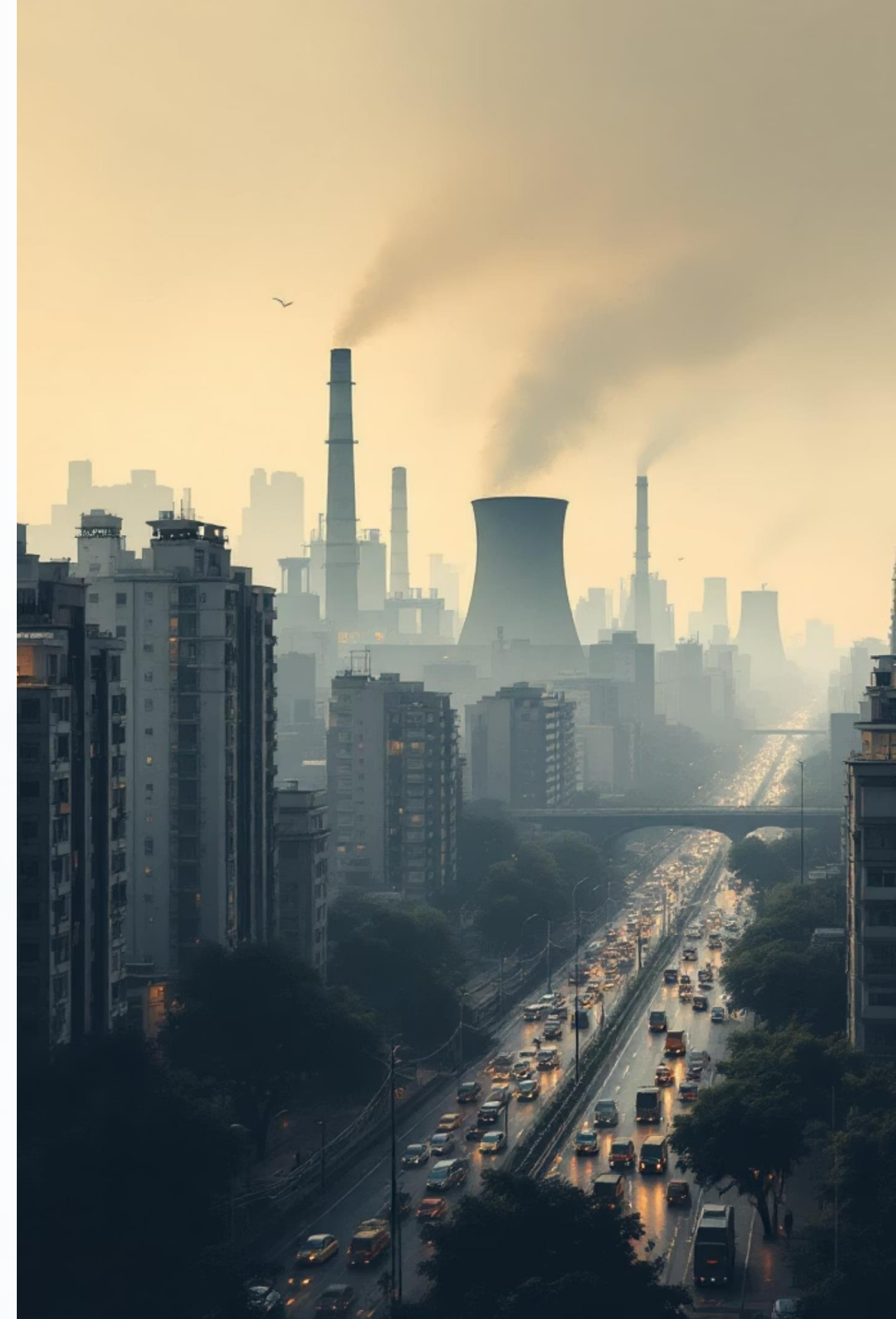
India is the world's third-largest energy consumer, with demand growing rapidly as industry, infrastructure, and a population of 1.4 billion expand.

🔥 CO_2 Emissions

Over 55% of India's electricity still comes from coal, making it one of the largest emitters of carbon dioxide globally.

🌱 The Sustainable Imperative

Air pollution, climate change, and import costs make a compelling case for transitioning to clean, indigenous renewable energy sources.





DEFINITION

What is Renewable Energy?

Renewable energy comes from **naturally replenished sources** — resources that are constantly restored by nature and will never run out on any human timescale.

→ Naturally Replenished

Sunlight, wind, rain, tides, and geothermal heat are continuously available.

→ Low Carbon Footprint

Unlike coal or oil, renewables emit little to no greenhouse gases during operation.

→ Inexhaustible vs Finite

Fossil fuels take millions of years to form; renewables regenerate in days, seasons, or continuously.

Common Sources of Renewable Energy

Six primary renewable sources power India's clean energy ambitions, each harnessing a different natural force.



सौर ऊर्जा

Converts sunlight into electricity using photovoltaic panels or concentrated solar thermal systems.



Wind Energy

Harnesses kinetic energy from moving air using turbines, particularly strong along India's coastlines and Tamil Nadu plateau.



Hydropower

Uses the energy of flowing or falling water to generate electricity; India's oldest and largest renewable source.



Geothermal

Taps heat from the Earth's interior; India has potential in the Himalayan region and Deccan volcanic belt.



बायो गैस

Converts organic waste — agricultural residue, wood, and biogas — into heat and electricity.



Ocean Energy

Captures energy from tidal movements and waves; India's 7,500 km coastline offers significant untapped potential.

Renewable Energy in India

4th

Global Wind Power

India ranks 4th globally in installed wind energy capacity.

DPO

5th largest solar power producer in the world.

India is the 5th largest solar power producer in the world.

500GW

2030 Target

India aims for 500 GW of non-fossil fuel capacity by 2030 under its NDC commitments.

- ❏ **National Solar Mission (2010):** A landmark government initiative targeting affordable solar energy across India, now part of the broader **National Action Plan on Climate Change (NAPCC)**. PM-KUSUM and PLI schemes further accelerate domestic manufacturing and rural solar deployment.





CASE STUDY

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Bhadla, Jodhpur district, Rajasthan — a sun-drenched desert region with over 320 sunny days per year.

⚡ Capacity

~2,245 MW installed capacity — one of the **largest solar parks in the world**.

🏠 Impact

Powers millions of homes across Rajasthan, reducing dependence on coal-based thermal plants.

🌍 Significance

A global model for utility-scale solar deployment; a cornerstone of India's clean energy future.

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✓ Advantages

□ Eco-Friendly

Minimal greenhouse gas emissions during operation, helping India meet its Paris Agreement commitments.

□ Sustainable & Inexhaustible

Sun, wind, and water are perpetually available, ensuring long-term energy security without depletion.

□ Reduces Pollution

Dramatically cuts air and water pollution, directly improving public health in India's cities.

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□ High Initial Cost

Infrastructure investment for solar parks, wind farms, and grid upgrades requires significant upfront capital.

□ Weather Dependent

Solar and wind output varies with weather conditions, making consistent baseload generation difficult.

□ Storage Limitations

Battery and grid storage technology must advance to effectively manage intermittent renewable supply.



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Renewable energy is not merely an option — it is India's most viable path to energy security, economic resilience, and a sustainable climate future. With bold government targets, a vast resource base, and growing investment, India is poised to lead the global clean energy transition.



Sustainability

Securing clean energy for generations to come.



Economy

Creating millions of green jobs and reducing import bills.



Climate

Meeting India's net-zero commitments by 2070.